

Lab

Java basics — Vars and selection

Lab: Using conditionals

Objective

In this lab, you'll practice using the logical AND (&&) and OR (||) in if statements.

Instructions

In this part you'll use if statements to grade a mark for an exam.

1. Back in the **labs** project, add a new class called **Program** in the package named **lab2**.
Make sure the new class has a **main** method.
2. Write code in the main method that implements the following grading rules and displays the grade:

If the mark is less than 1 or greater than 100, display:

*"Error: **marks must be between 1..100**"*

Less than 50	Fail
Between 50..60 (inclusive)	Pass
Between 61..70 (inclusive)	Merit
Between 71..100 (inclusive)	Distinction

Lab — optional

Airport Currency Exchange and Tax Calculator

Objectives

By the end of this lab, you will be able to:

- Use if statements for conditionals.
- Use switch to select between options.
- Declare and use variables of different types.
- Apply casting between double, int, and float.

Scenario

You've been hired to write a Java program for an **airport kiosk**. The kiosk helps travellers:

1. Convert **GBP** to one of three foreign currencies.
2. Apply an **airport service fee** based on the amount exchanged.
3. Calculate **airport tax** depending on customer type.

Requirements

1. The user has:
 - Entered the amount in **GBP** they want to exchange.
 - Has chosen a **currency** (1 = USD, 2 = EUR, 3 = JPY).
 - Entered the **customer type** (R = Regular, V = VIP).
2. Use a switch statement to:
 - Convert GBP to the chosen currency using fixed rates.
 - USD: 1 GBP = 1.25 USD
 - EUR: 1 GBP = 1.15 EUR
 - JPY: 1 GBP = 180.50 JPY
3. Apply a **service fee** using an if statement:
 - If amount < £100 → 5% fee
 - If amount >= £100 → 2.5% fee
4. Apply **airport tax** based on customer type:

- Regular (R) → 10% tax
 - VIP (V) → 5% tax
5. Display:
- Amount after service fee.
 - Amount after tax.
 - Final amount in foreign currency (cast to int for whole units).
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Starter Code Skeleton

There is no starter code. Please create a new project with a class called **Airport** in a package named **'qa'** that has a `main()` method.;

Your tasks

1. Complete and run your code for different amount of money and types of currency..
2. Modify the program to include:
 - Additional customer type: Staff (S) → 0% tax.
 - If you have used iteration in Java:
Option to repeat the exchange using a loop.
3. Challenge: Add more currencies and make exchange rates dynamic using float.



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